



ENHANCED ENTITY-RELATIONSHIP P-MODELING

(EER)

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In addition to ER model concepts
EE-R includes –

- ▶ Subclasses and Super classes.
- ▶ Specialization and Generalization.
- ▶ Category or union type.

Enhanced ER Model or Extended Entity-Relationship (EE-R) Model

- ▶ EER is a high-level data model that incorporates the extensions to the original ER model.
- ▶ Enhanced ERD are high level models that represent the requirements and complexities of complex database.
- ▶ Today the complexity of the data is increasing so it becomes more and more difficult to use the traditional ER model for database modeling.
- ▶ To reduce this complexity of modeling we have to make improvements or enhancements to the existing ER model to make it able to handle the complex application in a better way.

Enhanced entity-relationship diagrams

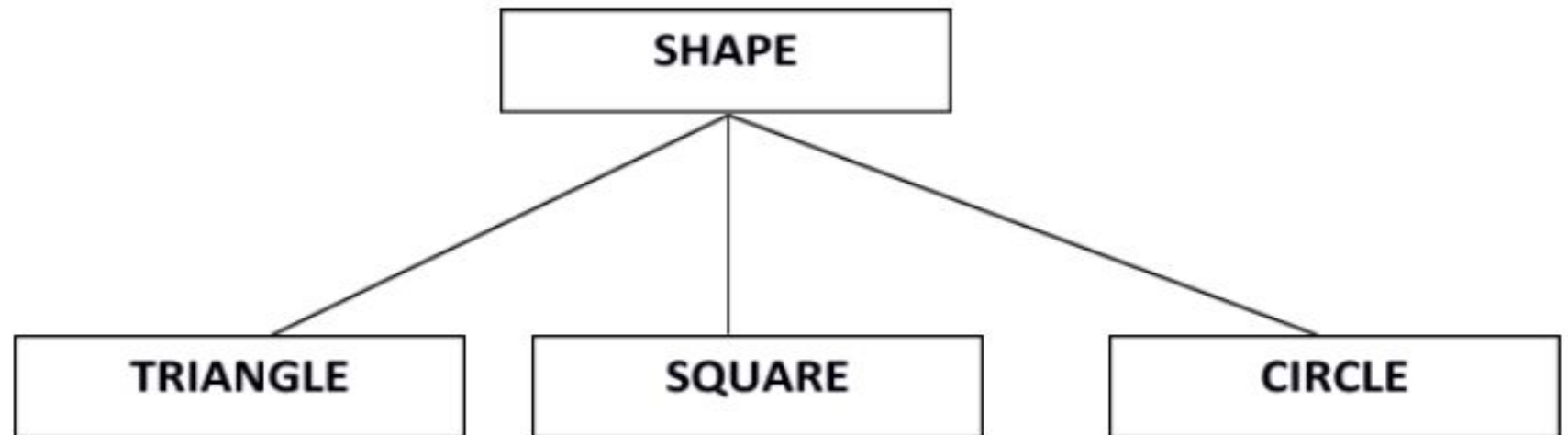
- ▶ Enhanced entity-relationship diagrams are advanced database diagrams very similar to regular ER diagrams which represent the requirements and complexities of complex databases.
- ▶ It is a diagrammatic technique for displaying the Sub Class and Super Class; Specialization and Generalization; Union or Category; Aggregation etc.
- ▶ **In addition to ER model concepts EE-R includes –**
- ▶ **Subclasses and Super classes.**
- ▶ **Specialization and Generalization.**
- ▶ **Category or union type.**

Features of the EER Model

- ▶ EER generates designs that are more accurate to database schemas.
- ▶ The diagrammatic style is useful in displaying the EER schema.
- ▶ The modeling concepts that are included in the ER model are also included in the EER model.
- ▶ The data properties and constraints reflected by the EER model are very precise.
- ▶ It incorporates the ideas of specialization and generalization.
- ▶ It is used to represent a collection of objects that are a union of distinct types of objects.

Subclasses and Super class

- ▶ Super class is an entity that can be divided into further subtype.
- ▶ consider Shape super class.
- ▶ Super class shape has sub groups: Triangle, Square and Circle.
- ▶ Sub classes are the group of entities with some unique attributes. Sub class inherits the properties and attributes from super class.



Subclasses and Superclasses

- ▶ In Many cases an entity type has numerous subgrouping of its entities that are meaningful and need to be represented explicitly because of their significance to the database application.
- ▶ An entity type may have additional meaningful subgroupings of its entities
- ▶ For Example : EMPLOYEE may be further grouped into:
 - ▶ □ Example: SECRETARY,, ENGINEER, TECHNICIAN, ... Based on the EMPLOYEE's Job
 - ▶ □ MANAGER , EMPLOYEEs who are managers
 - ▶ □ SALARIED_EMPLOYEE, HOURLY_EMPLOYEE Based on the EMPLOYEE's method of pay
 - ▶ □ EER diagrams extend ER diagrams to represent these
- ▶ additional subgroupings, called **subclasses or subtypes**

Subclasses and Superclasses

- ▶ Each of these subgroupings is a subset of EMPLOYEE entities
- ▶ □ EMPLOYEE is the superclass for each of these
- ▶ **subclasses**
- ▶ □ These are called superclass/subclass relationships:
- ▶ □ EMPLOYEE/SECRETARY
- ▶ □ EMPLOYEE/TECHNICIAN
- ▶ □ EMPLOYEE/MANAGER

Subclasses and Superclasses (3)

- ▶ These are also called IS-A relationships
- ▶ SECRETARY IS-A EMPLOYEE, TECHNICIAN IS-A EMPLOYEE,
- ▶ Note: An entity that is member of a subclass represents the same real-world entity as some member of the superclass:
- ▶ The subclass member is the same entity in a distinct specific role
- ▶ A member of the superclass can be optionally included as a member of any number of its subclasses.

When to use Super/Subtype Relationships:-

- ▶ These Relationships are used either (or) both of the following conditions are present.
 1. There are attributes that belongs some (but not all) of the instances of an entity type.
 2. The instances of a subtype has a relationship unique to the subtype.

Generalization and Specialization

- ▶ These are two normal kinds of relationships that were added to the normal ER model for enhancement.
- ▶ These are inspired by the object-oriented paradigm, where we divide the code into classes and objects, and in the same way, we have divided entities into subclass and superclasses.
- ▶ Specialized classes are called subclasses, and generalized classes are called superclasses or base classes.
- ▶ We can learn the concept of subclass by 'IS-A' analysis. For example, 'Laptop IS-A computer.' Or 'Clerk IS-A employee.'
- ▶ For example, in a university, a faculty member or clerk is a specialized class of employees. So an employee is a generalized class, and all others are its subclass.

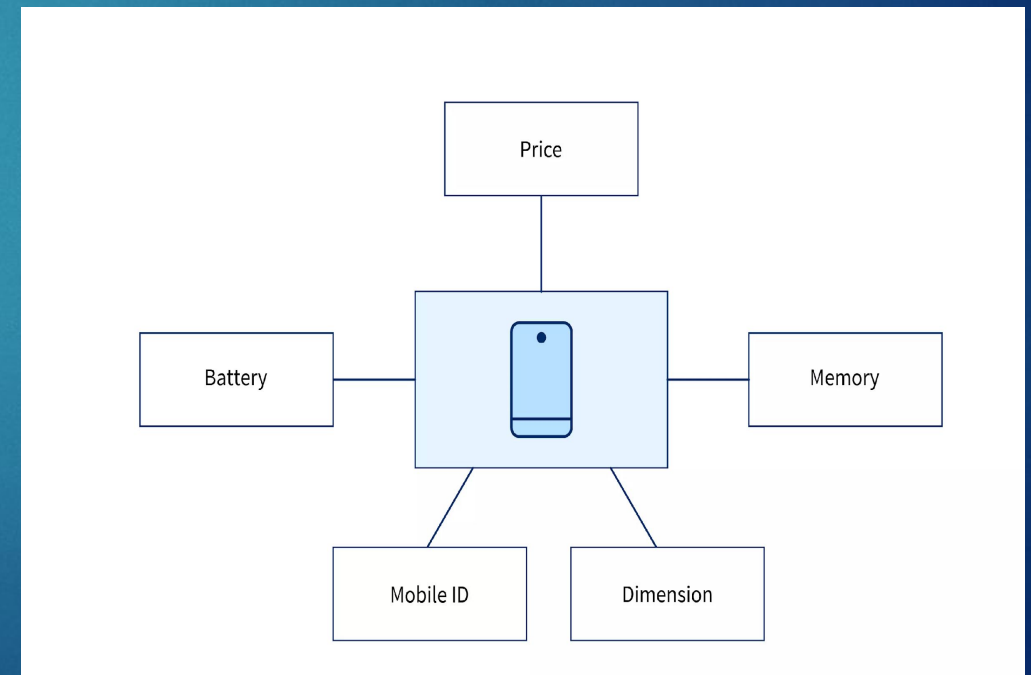
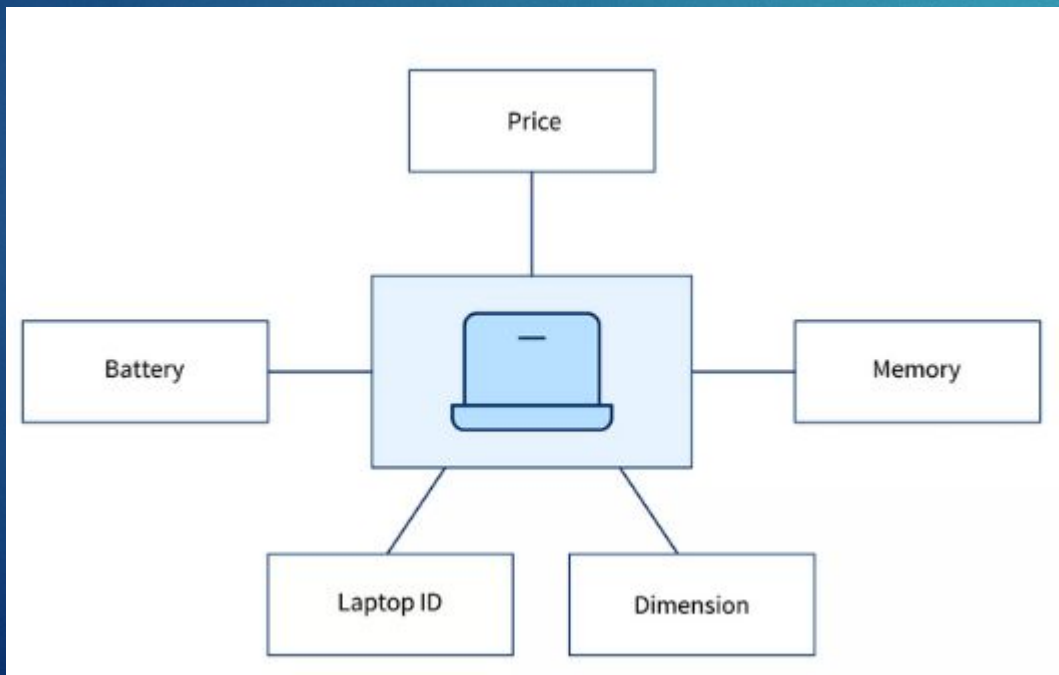
Generalization Characteristics



- ▶ we generalize and refer to them as "Social Media". This method of combining multiple lower-level entities of a similar kind and referring to them as a single higher-level entity is what we know as generalization.
- ▶ **Generalization Characteristics:**
 - ▶ The bottom-up technique is used for generalization.
 - ▶ It simplifies or generalizes the entities.
 - ▶ Higher-level entities and lower-level entities can be linked.
 - ▶ **Example:-**Let's say we have entities such as mobile and laptop and we need to generalize it based on the common entities between them so before generalization, we will have two separate entities for mobiles and laptops.

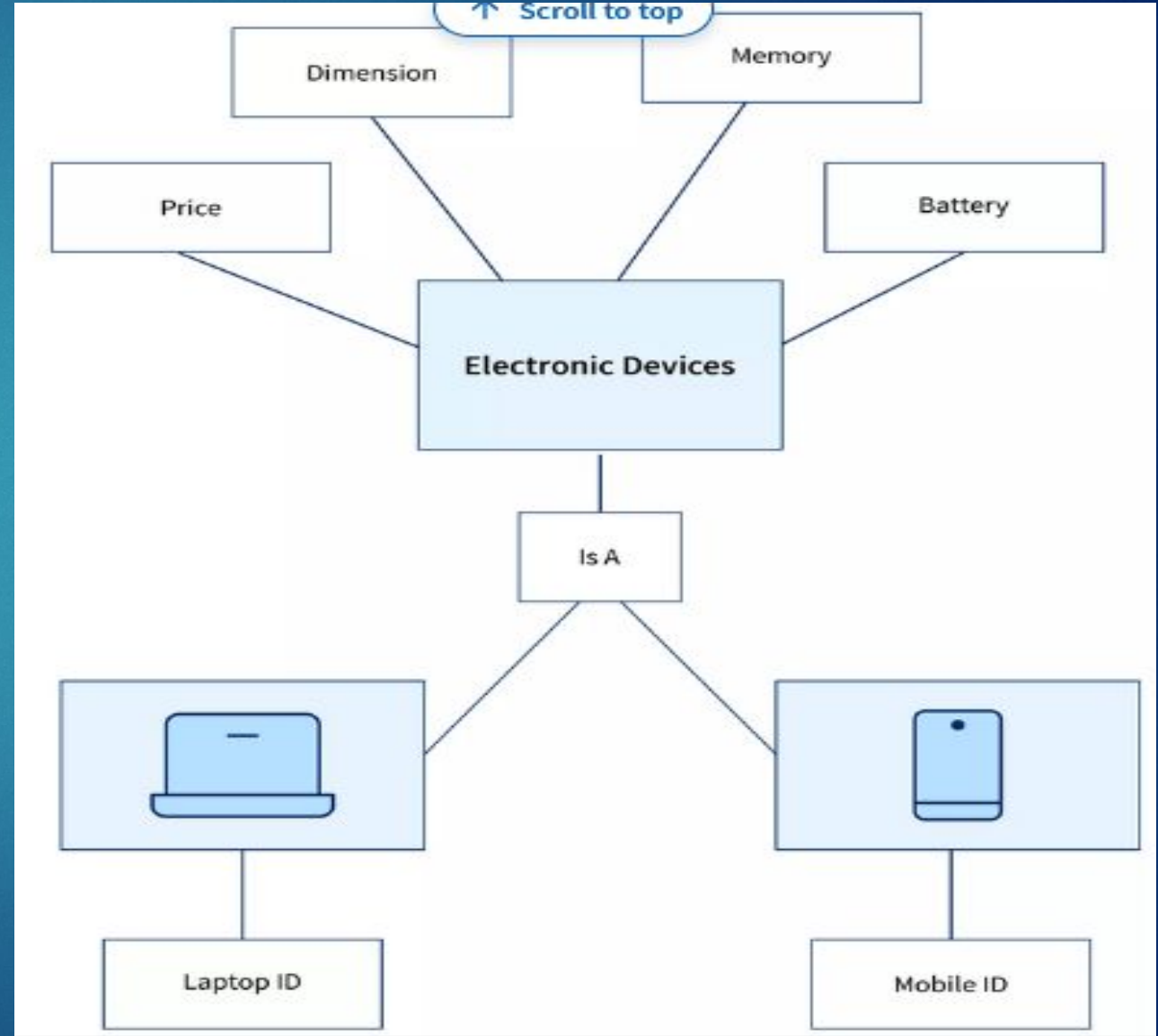
Entities before Generalization

- ▶ Below will be the entities before generalization:
- ▶ **Mobile Entity:** Mobile entity can have attributes like mobile_ID, price, memory, battery, and dimension to which we can apply the generalization procedure.
- ▶ **Laptop Entity:** Laptop entity can have attributes like laptop_ID, price, memory, battery, and dimension to which we can apply the generalization procedure.



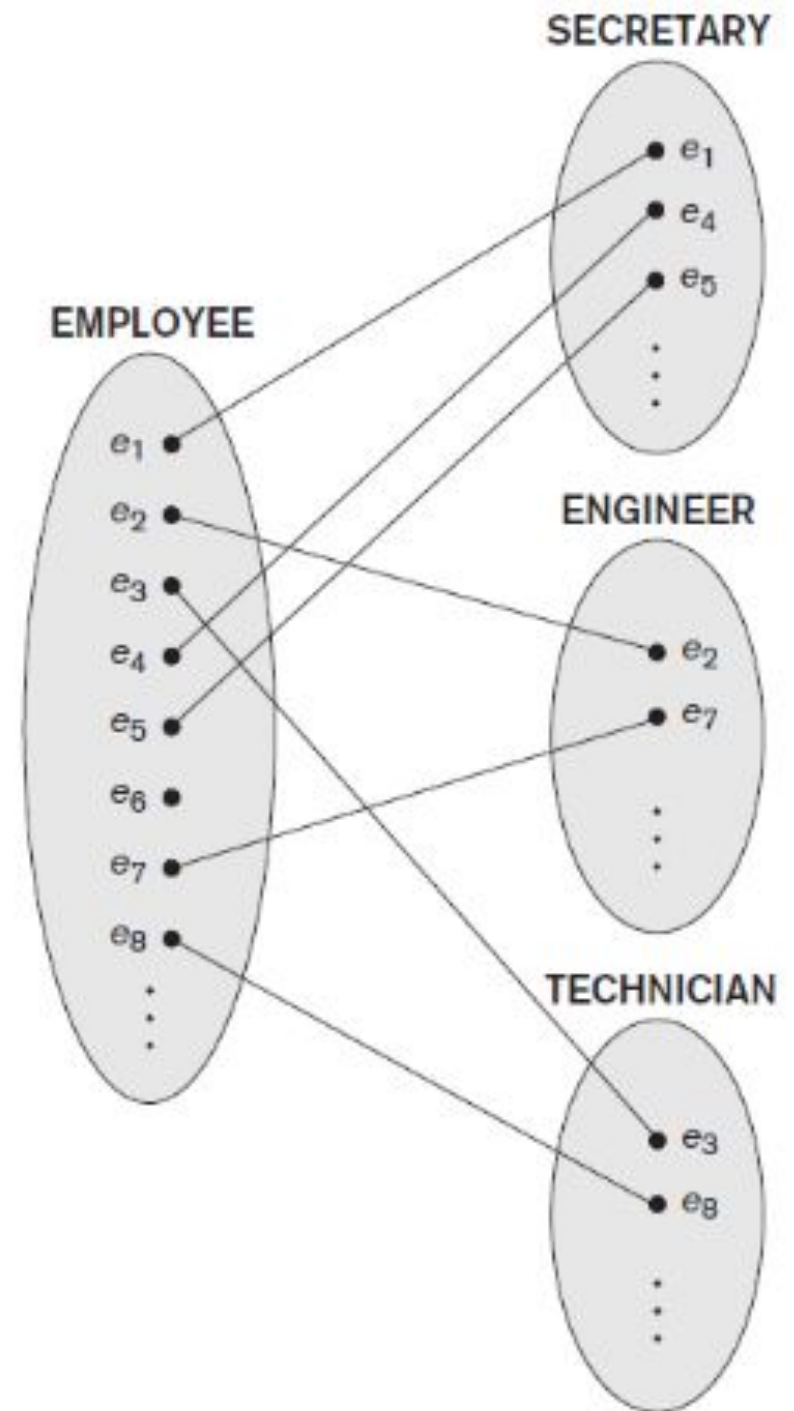
ER Diagram after Generalization

- ▶ We can see that for laptops and mobiles, some attributes like price, memory, battery, and dimension are the same so we can apply generalization and can group them into an entity called Electronic Devices.
- ▶ The Electronic Devices entity will have these common attributes and the laptop and mobile entity will be related to them using the "Is A" relationship table, so using generalization we can represent the ER diagram by:

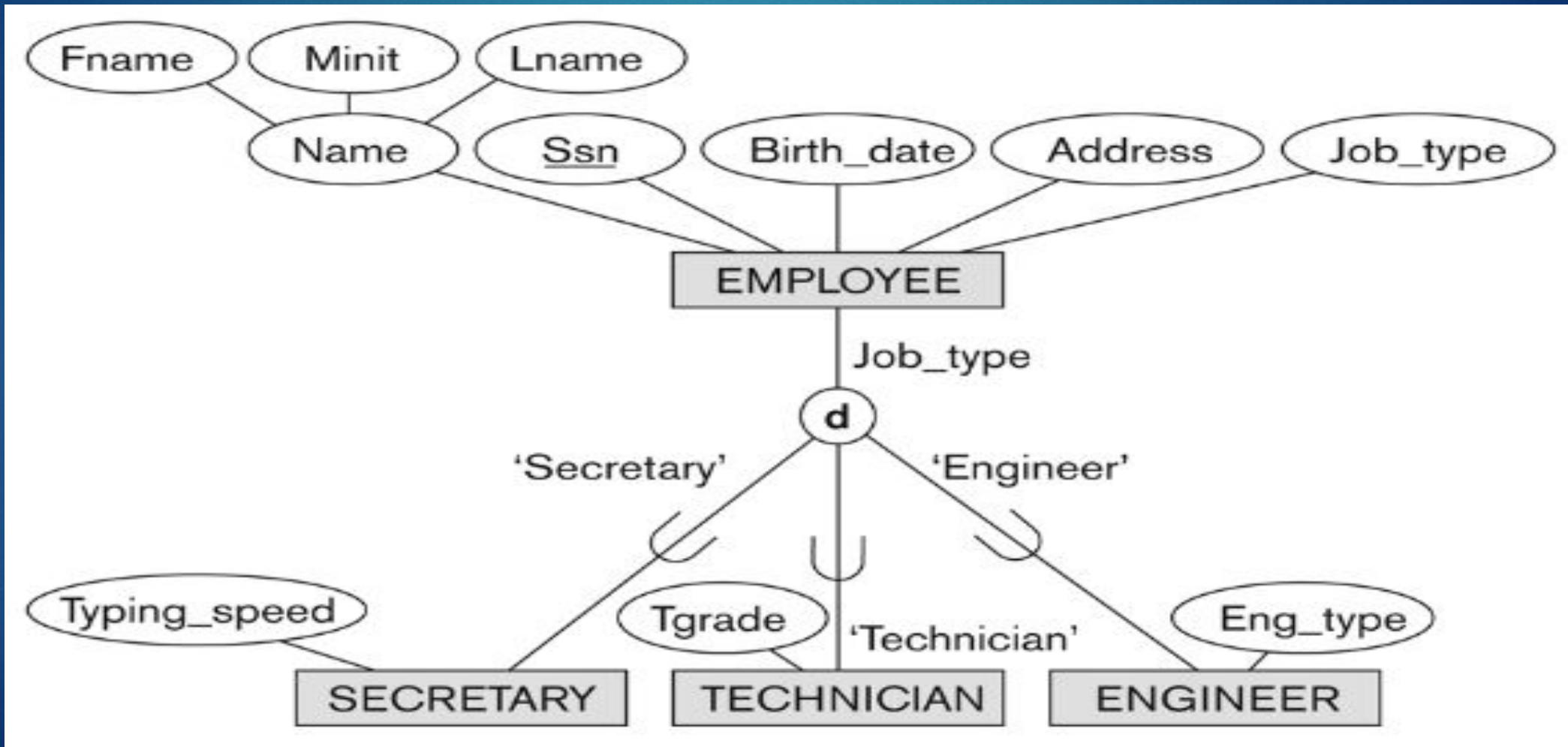


Specialization

- ▶ Process of defining a set of subclasses of an entity type
- ▶ Defined on the basis of some distinguishing characteristic of the entities in the superclass.
- ▶ Subclass can define:
 1. Specific attributes
 2. Specific relationship types



EER Diagram for attribute-defined specialization on Job_type



Example Specialization in DBMS

- ▶ A software developer is a person who can develop software. Now the software can be a web application, a mobile application, or, say, a desktop application.
- ▶ Hence, we can have developers specializing in a particular field, whether it's desktop applications, mobile applications, or desktop applications.
- ▶ This intuition of categorizing or dividing a higher-level entity into multiple lower-level entities of a similar kind is known as specialization.

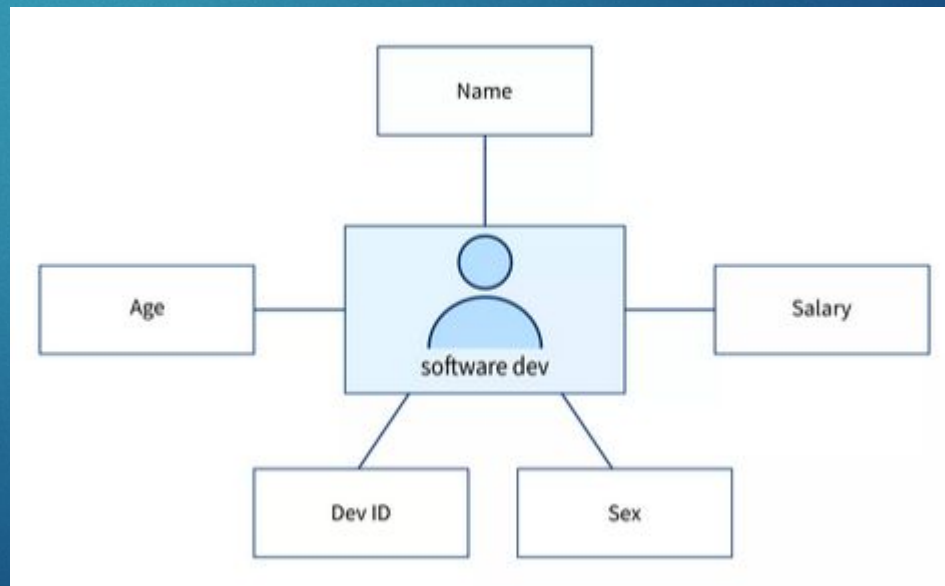
Specialization Characteristics:

The top-down technique is used for specialization. It divides or specializes in the entity.

Higher-level entities and lower-level entities can be linked.

Entities Before Specialization

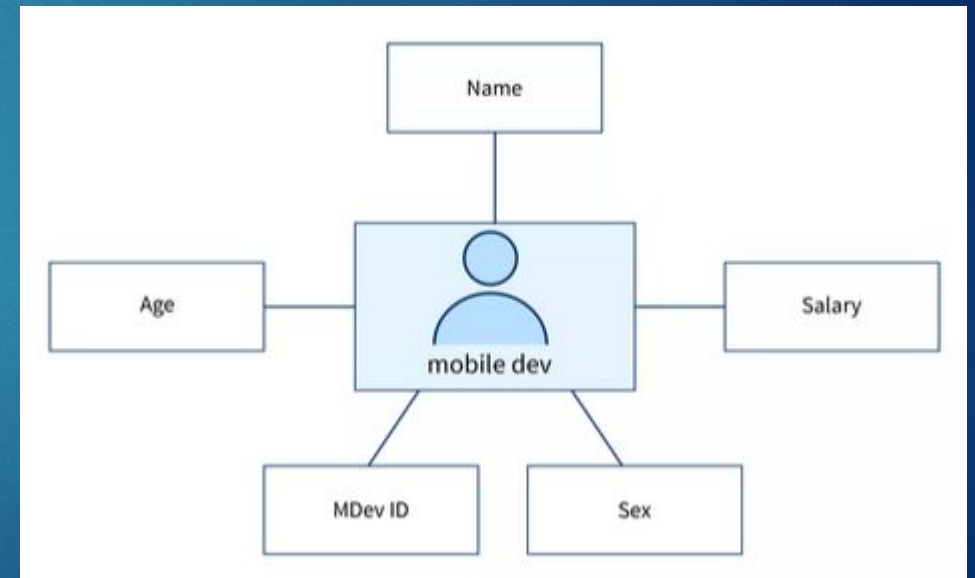
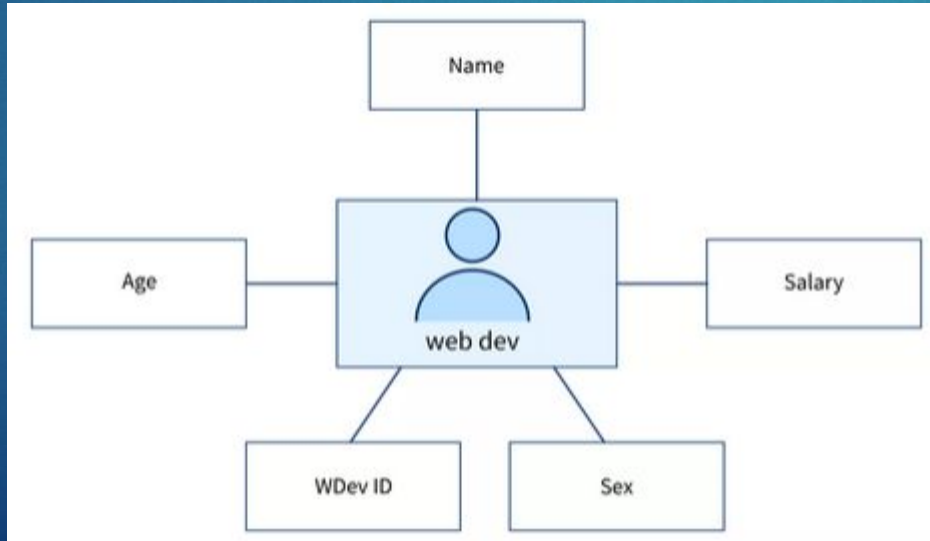
Software Developer: Software developers can have attributes like name, age, salary, etc. We will apply specialization to this entity and will divide it into entities.



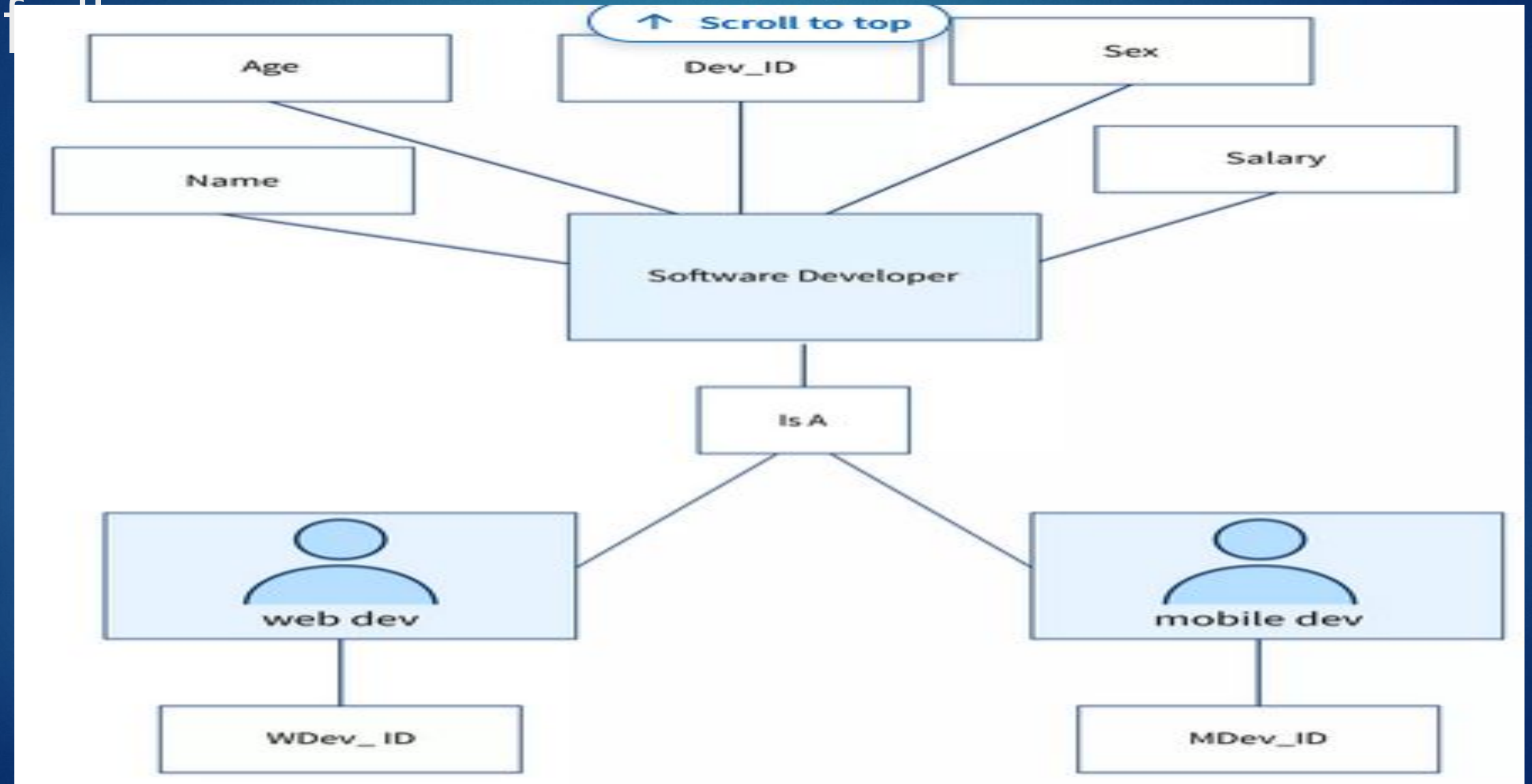
ER Diagram after Specialization

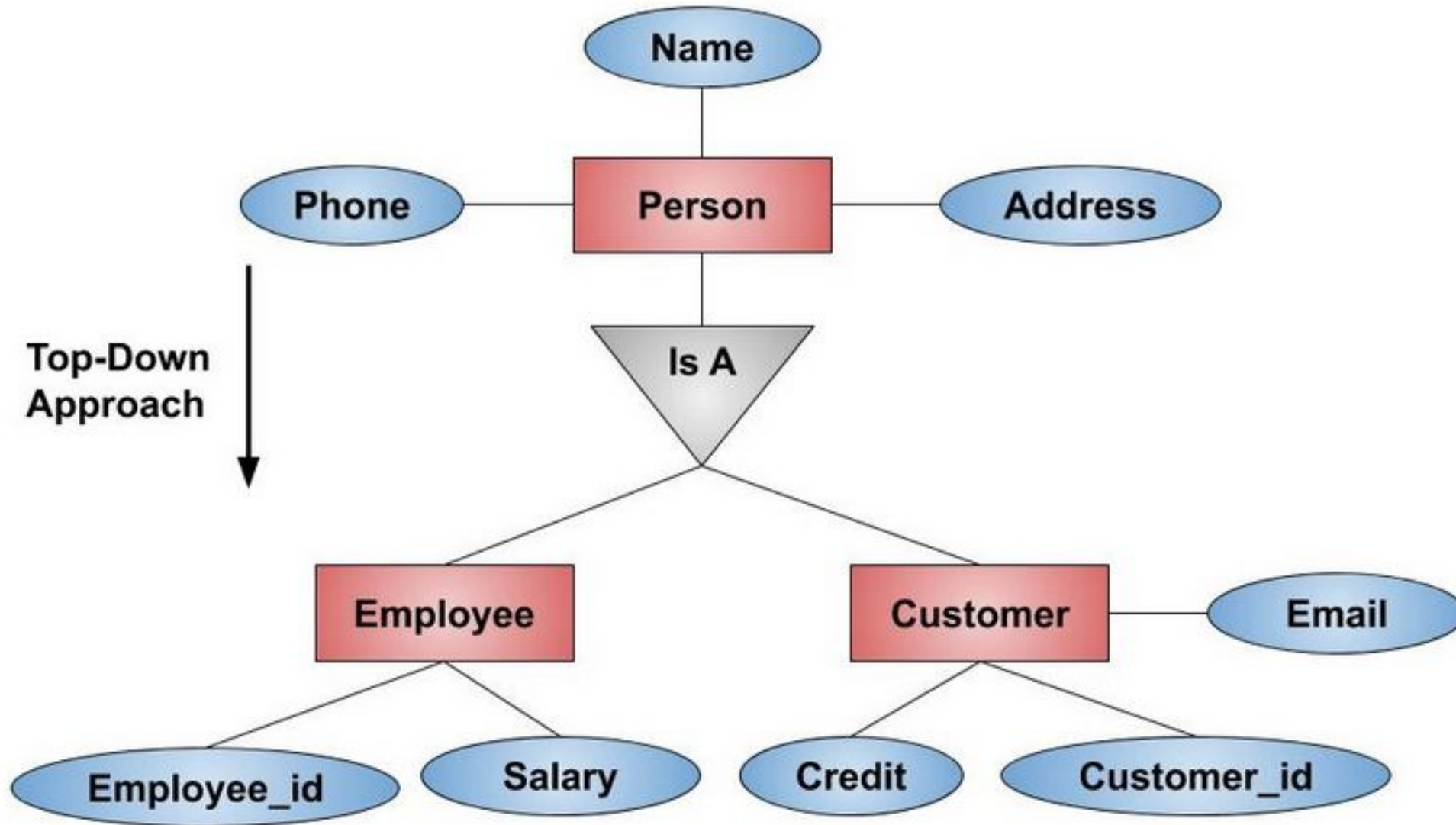
- ▶ We can specialize the software developer entity into the mobile developer entity and web developer entity.
- ▶ **Web Developer Entity:** A web Developer entity can have attributes like name, age, salary, and sex and is obtained after applying specialization in a software developer entity.

Mobile Developer Entity: Mobile Developer entity can have attributes like name, age, salary, and sex and is obtained after applying specialization in software developer entity.

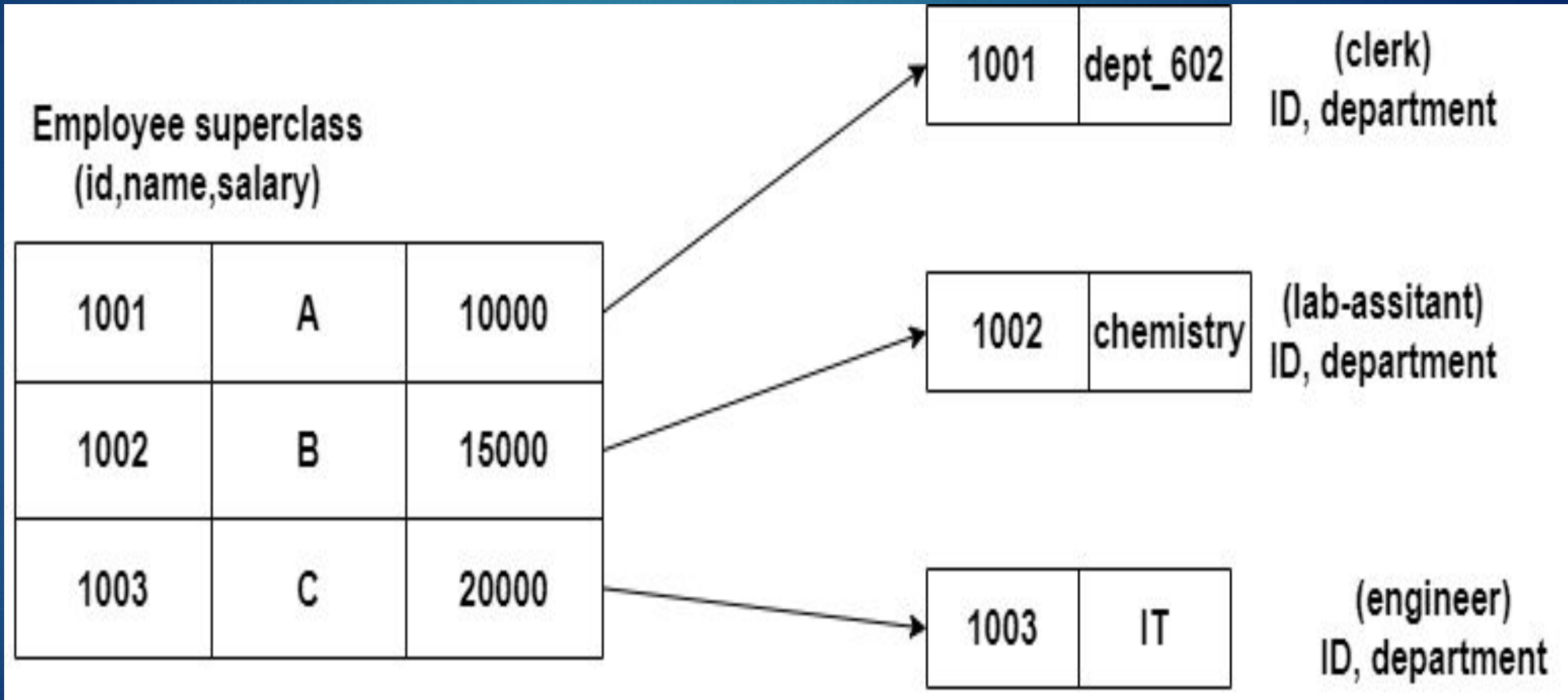


The ER diagram for the same is as



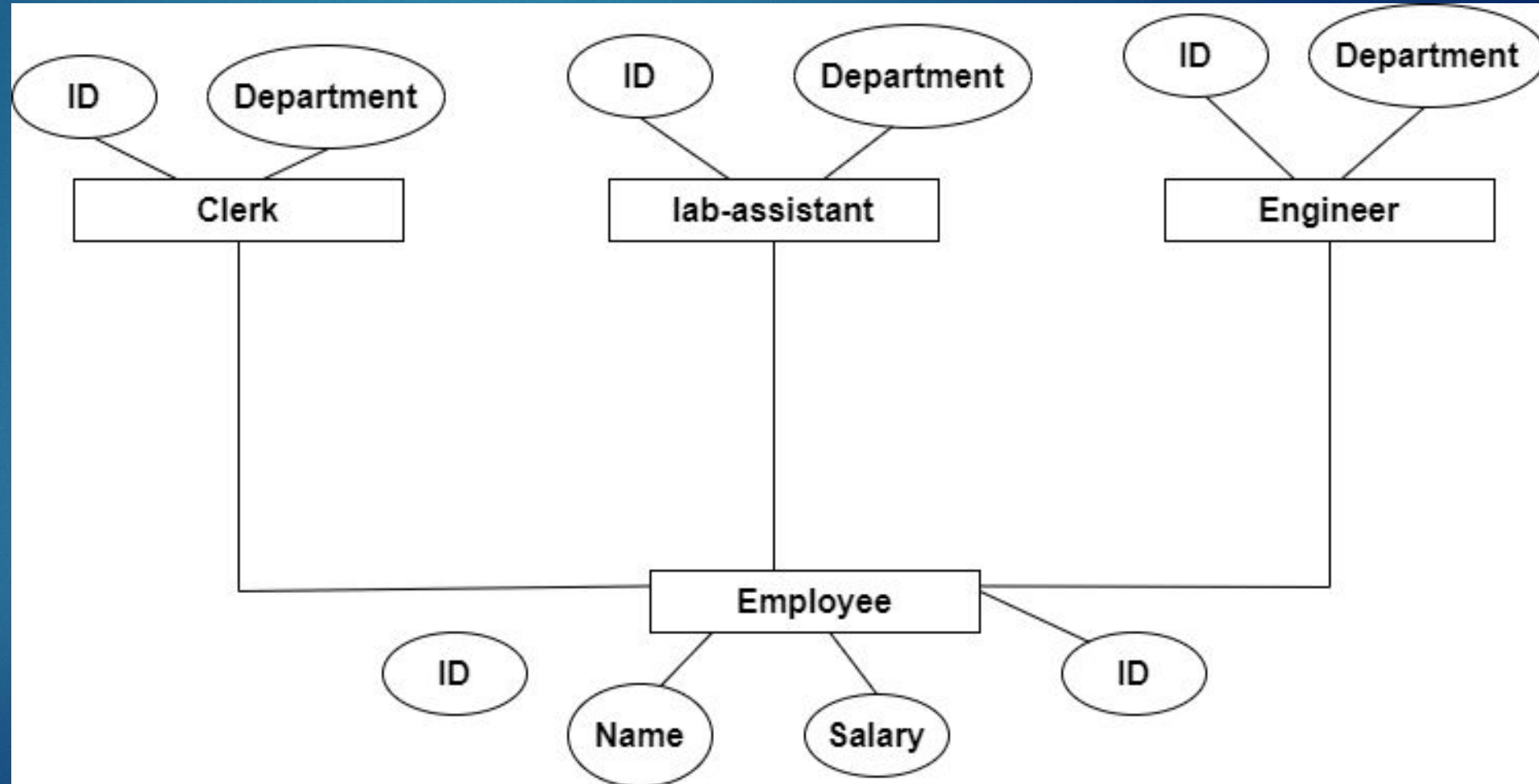


Let's suppose we have a superclass Employee and subclasses as a clerk, engineer, and lab assistant.



The Enhanced ER diagram of the above example will look like this:

- ▶ In the above example, we have one superclass and three subclasses. Each subclass inherits all the attributes from its superclass so that a lab assistant will have all its attributes, like its name, salary, etc.



Difference between Generalization and Specialization in DBMS

- ▶ The following are the **key difference** between generalization and specialization:



Generalization	Specialization
Bottom Up Approach	Top Down Approach
Schema Size Reduce	Schema Size Increases
Combining Entities	Splitting an Entities

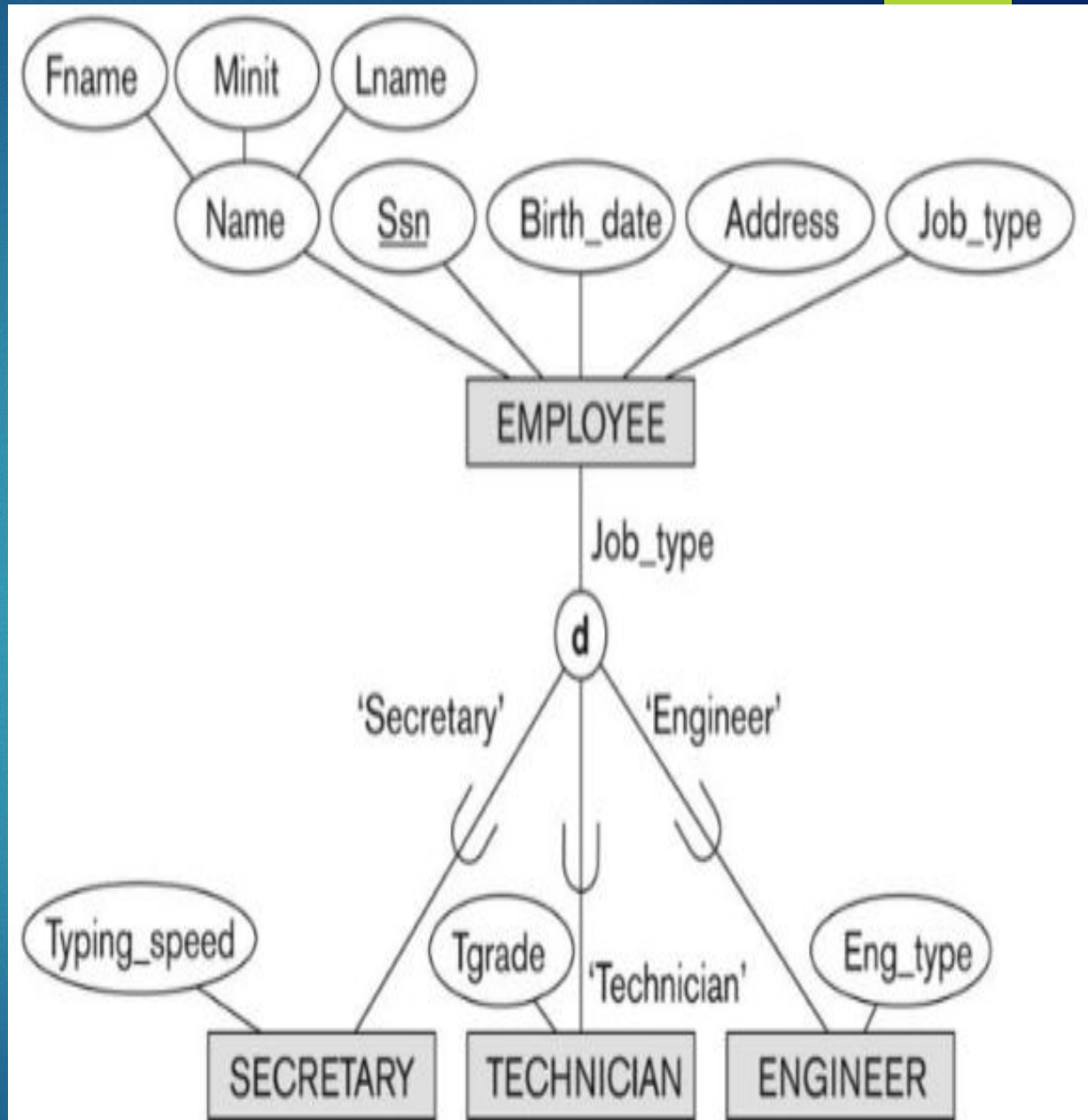
Constraints on Specialization and Generalization

- ▶ Two basic constraints can apply to a specialization/generalization:
- ▶ Disjointness Constraint
- ▶ Completeness Constraint.

Disjointness Constraint:

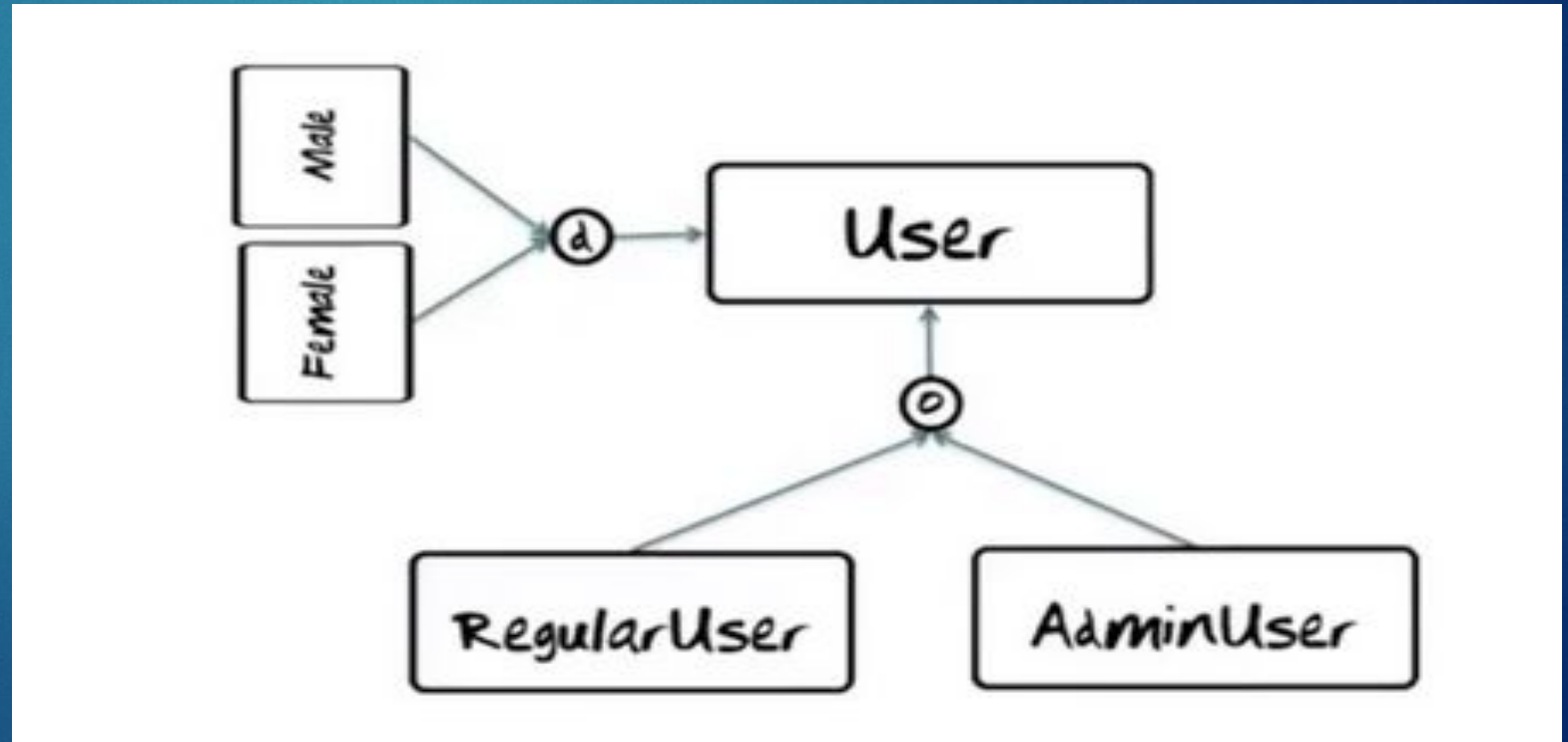
- Specifies that the subclasses of the specialization must be disjoint.
- An Entity can be a member of at most one of the subclasses of the specialization.

Specified by **'d'** in EER diagram.



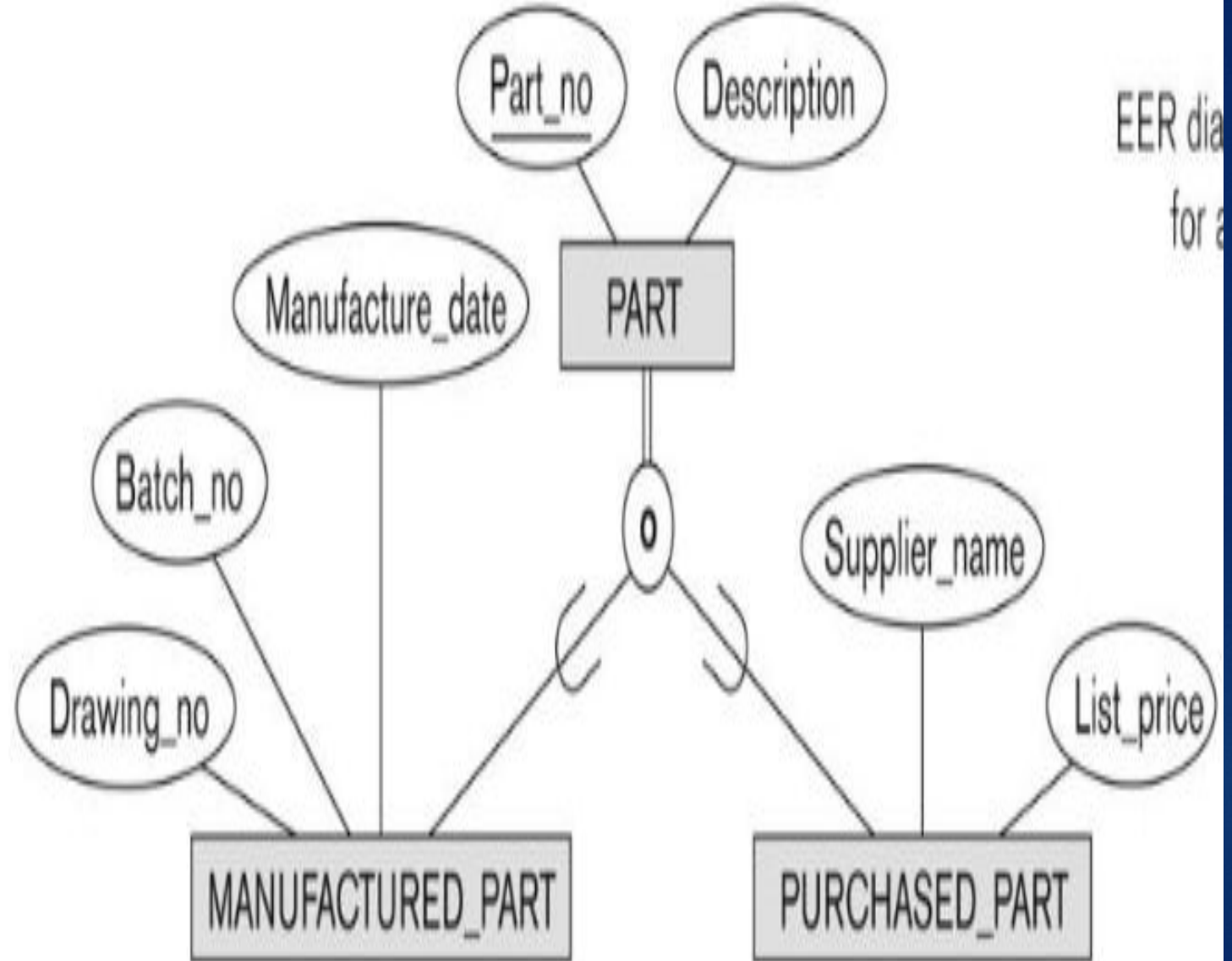
Constraints on EER

- ▶ Supertypes and Subtypes (“is-a” relationship types)
- ▶ **Disjointness Constraint** – the two subtypes cannot overlap. Indicated by the circled **d**.
- ▶ **Overlap Constraint** – the two subtypes can overlap. Indicated by the circled **o**.



overlapping

- ▶ If not disjoint, specialization is overlapping
- ▶ that is the same entity may be a member of more than one subclass of the specialization.
- ▶ Specified by 'o' in EER diagram



Completeness Constraint

- ▶ **Total** specifies that every entity in the superclass must be member of some subclass in the specialization/generalization. Show in EER diagrams by a **double line**
- ▶ **Partial** allows an entity not to belong to any of the subclasses. Shown in EER diagram by a **single Line**
- ▶ In general , a superclass that was identified through the generalization process usually **total**, because the superclass is derived from the subclasses and hence contains only the entities that are in the subclass.



Hence we have four types of Specialization/generalization

- ▶ Disjoint, total
- ▶ Disjoint, partial
- ▶ Overlapping ,total
- ▶ Overlapping ,partial

Some general rules:

- ▶ Deleting an entity from superclass implies that it is automatically deleted from all the subclasses to which it belongs.
- ▶ Inserting an entity in a superclass of a total specialization implies that the entity is mandatorily inserted in at least one of the subclasses of the specialization.

Lattice Example (UNIVERSITY)

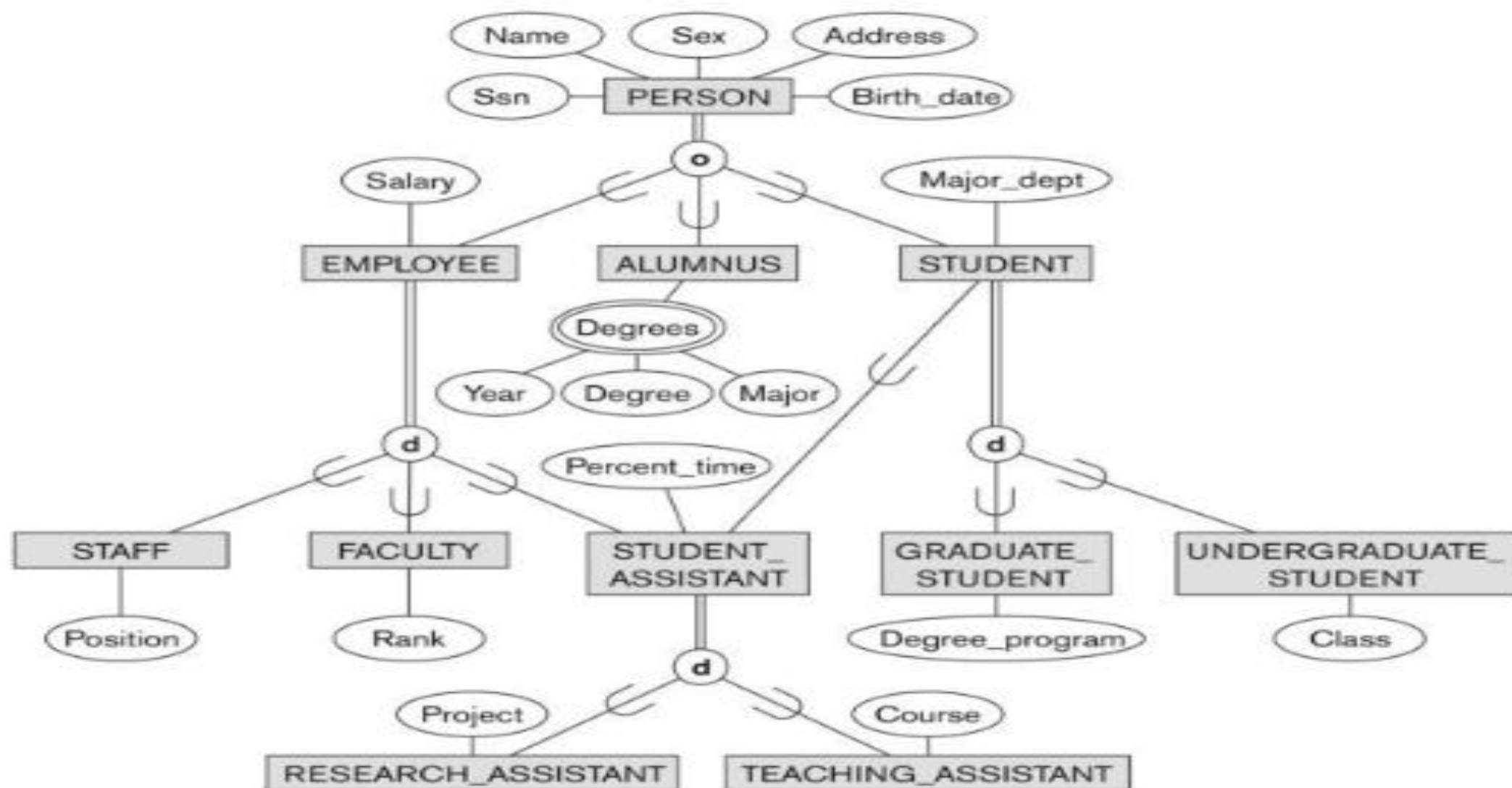


Figure 4.7

A specialization lattice with multiple inheritance for a UNIVERSITY database.

Union

- ▶ All of the superclass/subclass relationships we have seen so far origin from a single superclass.
- ▶ Some times we may need more than one superclass.
- ▶ In this case, the subclass will represent a collection of objects that is a subset of the UNION of distinct entity types.
- ▶ We call such subclass a **UNION TYPE**

Example: In a database for vehicle registration, a vehicle owner can be a PERSON, a BANK (holding a lien on a vehicle) or a COMPANY.

A UNION type called OWNER is created to represent a subset of the *Union* of the three superclasses COMPANY, BANK, and PERSON.

Two categories (UNION types):
OWNER,
REGISTERED
VEHICLE

Attribute inheritance works more selectively in the case of UNION.

For example, OWNER entity inherits attributes of a COMPANY, a PERSON OR a BANK

A shared subclass such as ENGINEERING_MANAGER inherits ALL the attributes of its superclasses

